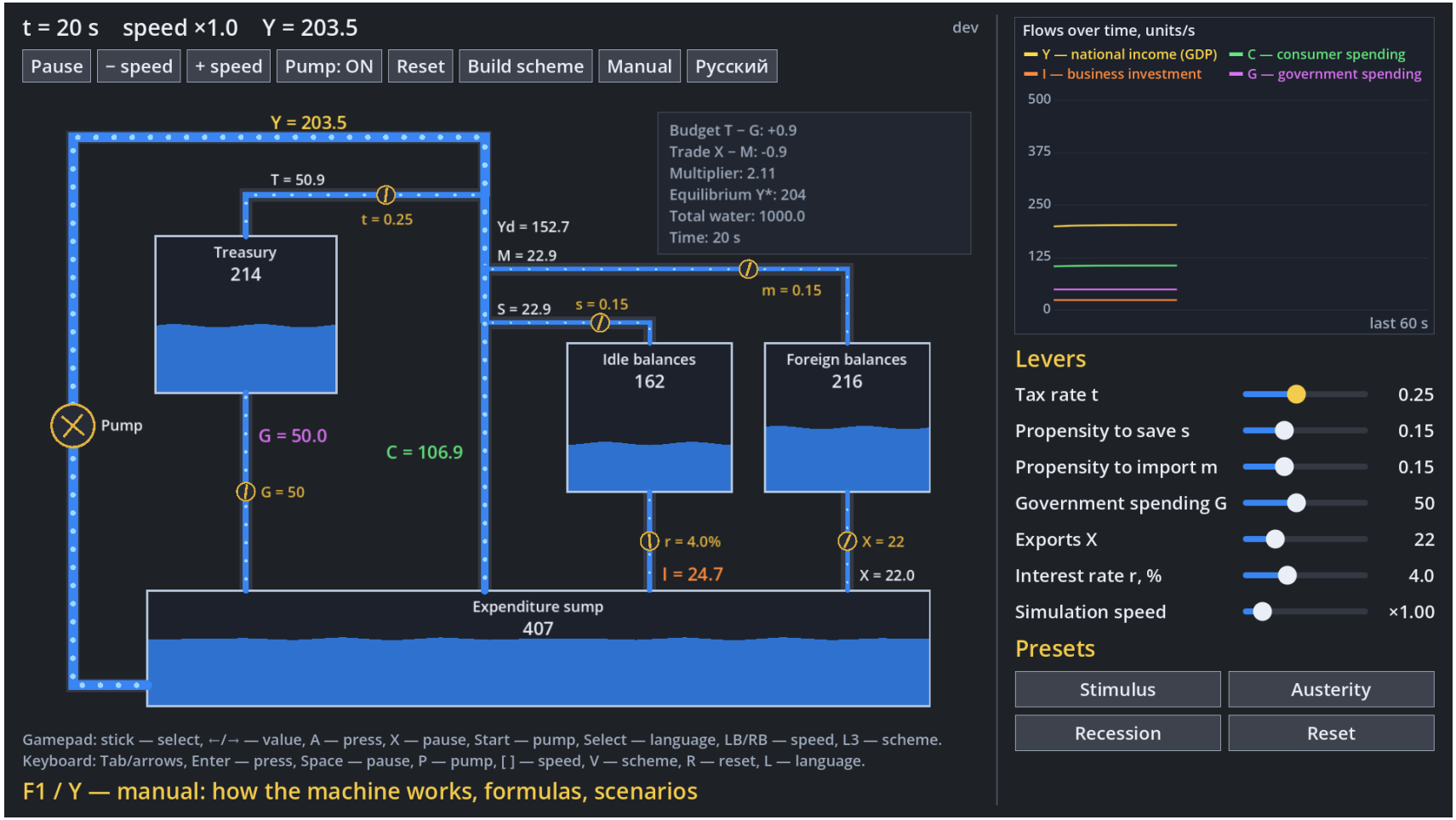
The Phillips Machine (MONIAC)

A hydraulic model of the economy: how it works, formulas, scenarios



The program screen. F1 or Y on the gamepad opens/closes this manual, ← / → or LB / RB flip pages.

1. What it is

In 1949 the New Zealand engineer and economist Bill Phillips, then a student at the London School of Economics, built together with Walter Newlyn a machine of perspex tanks, a pump and valves that showed a Keynesian model of the economy with water. Money is water: it circulates through pipes, collects in tanks and leaks out of circulation. The machine was called MONIAC (Monetary National Income Analogue Computer); copies were used for teaching at universities in several countries, and surviving machines are kept in museums, e.g. the Science Museum in London and the Reserve Bank of New Zealand.

The machine does not solve equations on paper — it lives them: turn a valve and every flow readjusts by itself. This program is a digital model of such a machine with simplified but fully open maths (section 3).

The key principle: water is never created or destroyed. Its total amount (1000 units) is conserved exactly — see the “Total water” line on screen.

2. Tanks and flows

Tank	Inflow	Outflow	Economic meaning
Expenditure sump (bottom)	C, G, I, X	pump → Y	Aggregate demand: everything spent on the country's goods and services
Treasury	T = t·Y	G — government spending valve	The budget: taxes come in, spending goes out. The level rises with a surplus and falls with a deficit
Idle balances	S = s·Yd	I — depends on the rate r	Money set aside by households; business takes it to invest
Foreign balances	M = m·Yd	X — exports valve	Money that went abroad for imports; it comes back when foreigners buy our exports

The pump lifts water from the bottom tank to the top — this is national income Y. At the top the flow splits: part goes to the Treasury (taxes T), the rest is disposable income Yd. Part of it goes to savings (S) and abroad (M), while consumption C drains straight back down. Government spending G, investment I and exports X return to the bottom too.

On the scheme: the thickness of water in a pipe and the speed of the moving dots show the flow; the valves (yellow circles) show the lever: the flap along the pipe means open, across means closed.

3. Model formulas

Quantity	Formula	Default
Income (pump output)	$Y = 0.5 \cdot \text{sump level}$	203.6
Taxes	$T = t \cdot Y$	$t = 0.25$
Disposable income	$Yd = Y - T$	
Saving	$S = s \cdot Yd$	$s = 0.15$
Imports	$M = m \cdot Yd$	$m = 0.15$
Consumption	$C = Yd - S - M$	
Government spending	G — set by the lever	50
Investment	$I = 45 \cdot e^{-0.15 \cdot r}$	$r = 4\ \% \rightarrow I = 24.7$
Exports	X — set by the lever	22

Each tank level changes as “inflow minus outflow”, e.g. for the Treasury d(level)/dt = T − G. While a tank is not empty its outflow is the set value; below 40 units the outflow is gradually choked — you cannot pour from an empty tank.

The multiplier

If no tank is empty, at equilibrium the pump lifts exactly as much as drains down: $Y = C + G + I + X$. Substituting $C = (1 - t)(1 - s - m) \cdot Y$ gives

$$\mathbf{Y^* = (G + I + X) \cdot k,\ k = 1 / (1 - (1 - t)(1 - s - m))}$$

With the default levers k = 2.11, Y* = 204: every extra unit of government spending ends up as 2.11 units of income. The multiplier and Y* are shown in the on-screen summary. The bigger the leakages (t, s, m), the smaller the multiplier.

Y does not jump to Y* at once: the time constant is about 1 / (0.5 · (1 − (1 − t)(1 − s − m))) ≈ 4.2 s of model time. And while Y = Y*, the tanks may slowly fill or drain — then the equilibrium moves (section 6).

4. Screen and controls

At the top: time, speed and Y, then the menu. Left: the machine; right: a chart of Y, C, I, G over the last 60 seconds, the levers and presets. The summary box on the scheme shows the budget T − G (plus = surplus), the trade balance X − M, the multiplier, the equilibrium Y*, total water and time.

Gamepad (Steam Deck)	Keyboard	Action
D-pad / left stick	arrows, Tab	move between buttons and sliders
← / → on a slider	← / →	change a lever (1/40 of the scale per press)
A	Enter	press a button
X	Space	pause / run
Y	F1	manual
Start	P	pump on / off
Select	L	language
LB / RB	[/]	speed ÷2 / ×2

Gamepad (Steam Deck)	Keyboard	Action
—	R	reset
in the manual: ← → LB RB, B	← →, PgUp/PgDn, Home, O, Esc	flip, first page, open the PDF, close

With a mouse you can click buttons and drag sliders. The language follows the system on first launch; the choice is remembered.

5. Presets

A preset changes only its own levers; water levels stay. In the table the start is after a reset and 60 s with the default levers (Y = 204), then the preset is applied; times are model time (speed ×1).

Preset	Levers	Y*	Y after 10 s	Y after 30 s	Y after 120 s	Treasury after 120 s
Stimulus	t = 0.18, G = 85, r = 1.5%	336	274	215	179	15
Austerity	t = 0.38, G = 20	118	110	63	53	878
Recession	s = 0.32, r = 9%, X = 8	116	120	61	56	11

Stimulus. Lower taxes, higher spending, cheap money — Y quickly rises towards the new Y*. But T − G turns negative: the Treasury drains, and once it is empty G falls to the level of taxes — the stimulus runs out. Watch the G line on the chart and the last column of the table.

Austerity. Higher taxes, lower spending: the budget is in surplus, water piles up in the Treasury — and drops out of circulation. Y and consumption fall.

Recession. Households save more, credit is expensive, exports have collapsed: leakages grew, less water comes back down — Y slumps.

Reset restores all levers, water levels, the speed and the chart.

6. Experiments

Experiment 1. The multiplier

1. Press Reset and wait 20–30 seconds for Y to settle.

2. Raise G by 10 (3–4 presses of → on the gamepad).

3. After 20 s Y grows by about 21 — close to k·10 = 21.

Experiment 2. The paradox of thrift

1. After a reset S ≈ 22.9, Y ≈ 204.

2. Raise s to 0.30 — everyone decides to save twice as much.

3. A minute later Y ≈ 125 and the saving flow S ≈ 28.1: income fell, and saving grew only a little, although everyone sets aside twice the share.

Experiment 3. A deficit does not last

1. After a reset set G = 120.

2. The Treasury is nearly empty in about 4 s. After that G falls to the tax income: 30 s later G ≈ 53 with T ≈ 53.

Experiment 4. Stop the pump

1. Turn the pump off (Start / P).

2. Income Y = 0 and the upper tanks drain down: almost all water (all but 10 units) is in the bottom tank in about 12 s. An economy without circulation stands still, although all the water is there.

This is a teaching model: the coefficients are chosen so that processes take seconds, not years, and the economy is nearly balanced with the default levers. The real MONIAC had more feedback loops — e.g. the interest rate depended on the idle-balances level; here r is set by a lever.

7. Problems: from a real situation to the machine's levers

Every problem starts from the same state: Reset, then a minute with the default levers (Y ≈ 204, Treasury ≈ 250, idle balances ≈ 91, foreign balances ≈ 252). Then the levers are changed, time is counted from that moment, speed ×1. All numbers come from the same model as the program — you can check them on screen.

General procedure. 1) Translate the event into levers: what changed — taxes, spending, the rate, propensities, exports. 2) Compute the equilibrium $Y^* = (G + I + X)\cdot k$ — the “short run” answer. 3) Watch which tanks drain or overfill meanwhile: they shift the equilibrium over time — the “long run” answer. The units are notional: directions and proportions matter, not absolute numbers.

Problem 1. A tax cut (USA, 1964)

Problem. In 1964 the USA cut income tax, following the Kennedy administration's plan, to speed up growth. Spending was not cut. How much will income grow? Does the cut pay for itself — will tax revenue rise? What happens to the Treasury?

Quantity	Calculation	On the machine
Change	Tax rate 0.25 → 0.20	slider t = 0.20
Multiplier	k = 1 / (1 − 0.80·0.70) = 2.27	“Multiplier” line
Equilibrium	Y* = (50 + 24.7 + 22)·k = 219.8	“Equilibrium Y*” line
Taxes	T = 0.20 · 219.8 = 44.0	was 50.9
Watch	Budget T − G and the Treasury level	summary, left tank

Solution. Income rises quickly: Y = 218.0 after 10 s and 219.6 by 20 s, almost Y* = 219.8. But taxes fall from 50.9 to 43.9: income growth does not make up for the lower rate, the budget goes into deficit T − G = -6.1.

The deficit is paid from the Treasury: its level falls from 250 to 54 in 30 s. By about second 60 the tank is nearly empty, the G valve is choked down to tax income and income falls: Y = 195.8 after a minute and 183.3 after 5 minutes, **below the start**.

Conclusion: a tax cut boosts income while there is money to pay for it. Without spending cuts or borrowing (the machine has none — only the Treasury's reserve) the effect is temporary.

Problem 2. An export boom (New Zealand, 1950–1951)

Problem. During the Korean War world wool prices — the main export of New Zealand, Phillips's home country — rose several times over. Export earnings jumped. How does income change, and for how long?

Quantity	Calculation	On the machine
Change	Exports 22 → 40	slider X = 40
Equilibrium	$Y^* = (50 + 24.7 + 40)\cdot 2.11 = 241.5$	+37.9
Imports	$M = m\cdot(1 - t)\cdot Y^* = 27.2$	less than X — trade surplus
Watch	Foreign balances level, X − M	right tank, summary

Solution. Income rises fast: the peak Y ≈ 238.0 at about second 10. But water for exports comes from the foreign balances tank — money foreigners collected by selling us imports. X = 40 with M ≈ 26.8: the tank drops from 252 to 26 in 20 s.

Once the tank is nearly empty, exports are limited by imports: X = M ≈ 23.2 after a minute. Income comes back down: 209.4 after 30 s, 200.1 after 2 minutes. But the boom left a mark in the Treasury: riding on high incomes it grew from 250 to 541.

Conclusion: a commodity boom gives fast but temporary growth; a trade surplus lasts only while the rest of the world has money to pay. It is wise to put part of the windfall aside — as the Treasury did here.

Problem 3. Tight money (the Fed, 1979–1981)

Problem. To break inflation, the US Federal Reserve under Paul Volcker raised rates to nearly 20%. What happens to investment and income? Where does the money go?

Quantity	Calculation	On the machine
Change	Rate 4% → 12% (top of the scale)	slider r = 12
Investment	$I = 45 \cdot e^{-0.15 \cdot 12} = 7.4$	was 24.7
Equilibrium	$Y^* = (50 + 7.4 + 22) \cdot 2.11 = 167.2$	-36.3
Watch	Idle balances tank: S versus I	middle tank

Solution. Income falls to Y* in ~20 s: Y = 167.5. But saving S ≈ 18.8 now exceeds investment I = 7.4: the difference settles in the idle balances tank. It grows from 91 to 335 in 20 s and 707 in a minute.

Water stuck in idle balances has left circulation. The Treasury (in deficit because taxes fell) and foreign balances drain, and income slumps below the equilibrium: 121.5 after a minute, 66.1 after 5 minutes.

Conclusion: a high rate does not just cut investment — it makes money sit idle. This is how Volcker beat inflation at the cost of the deep 1981–1982 recession. Set r back to 4% and watch the stored water flow back into circulation.

Problem 4. The Great Depression and the New Deal (1930s)

Problem. After the 1929 crash people and banks hoard, credit is expensive, world trade has collapsed (the Recession preset). What government spending would restore the previous income? Is the Treasury enough? Does cheap credit help?

Quantity	Calculation	On the machine
Slump	s = 0.32, r = 9%, X = 8	Recession preset: Y* = 115.6
Multiplier	k = 1 / (1 − 0.75·0.53) = 1.66	leakages grew — k is smaller
Required G	G = 204 / 1.66 − 11.7 − 8 = 103	slider G = 103
Watch	Treasury, idle and foreign balances	all three tanks

Solution. Without action income falls to 80.0 in 20 s and 55.8 in a minute — below Y* = 115.6, because the Treasury drains fast and water piles up in idle balances (558).

With G = 103 the equilibrium returns to 204, but the deficit is huge and the Treasury is empty within seconds (20 left after 5 s). Income only gets up to 191.1, and is back to 57.8 after 30 s.

Combined with cheap credit (r = 0%, G = 80) water flows from idle balances into investment: Y = 204.3 after 10 s. But while exports are 8 and imports are larger, water leaks abroad (foreign balances 784 after a minute) and income slides again: 92.2.

Conclusion: in a depression no single measure works — all the leaks must be closed at once: spending, the rate and trade. Try to find a set of levers that keeps Y near 200 for at least 2 minutes.

Problem 5. A balanced budget (Haavelmo, 1945)

Problem. The Norwegian economist Trygve Haavelmo showed that if the state spends more and collects exactly as much in taxes, income still grows. Raise G by 20 and pick the tax rate t so that the budget stays balanced (T = G) at the new equilibrium. How much does income grow?

Quantity	Calculation	On the machine
Change	G = 50 → 70	slider G = 70
Condition	t · Y*(t) = G	adjust t until T − G ≈ 0
Answer for t	t ≈ 0.31 (found by trial)	slider t = 0.31
Equilibrium	Y* = 225.7	+22.1

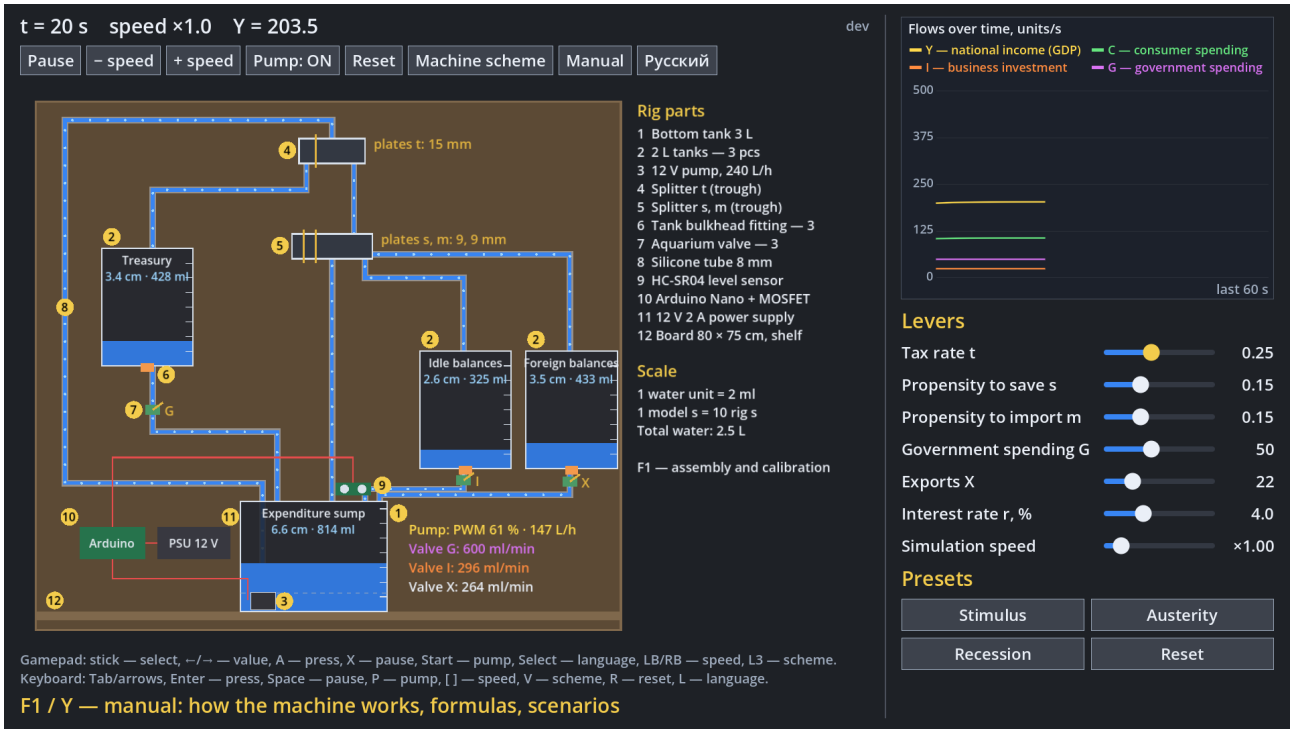
Solution. After 20 s Y = 225.6, budget T − G = -0.1: practically zero. Income grew by 22.0 — almost exactly the extra spending (20). The balanced-budget multiplier is close to one.

Why: each unit of tax reduces consumption only by the share (1 − s − m), while each unit of spending is spent in full. The difference drives the multiplier.

This is the most stable of the five scenarios: Y = 222.5 after 2 minutes. The slow drift later (212.8 after 5 minutes) is because t is rounded to hundredths and the budget runs a small deficit, while imports are slightly above exports: the Treasury slowly drains (138 after 2 minutes) and foreign balances fill up.

8. Building the machine at home

The “Build scheme” button (V, L3 on the gamepad) shows the rig on a board: parts are numbered, and the same levers are shown in physical units: splitter plate positions in mm, valve flows in ml/min, the pump PWM, levels in cm and ml. The model is the same, so you can set up the rig straight from the screen.



Scale

Model	Rig	Example
1 water unit	2 ml	all water 1000 units = 2.0 L
1 model second	10 s	a 30 s experiment = 5 minutes on the rig
Flow 1 unit/s	12 ml/min	Y = 200 → 2.4 L/min = 144 L/h
Level in a 2 L tank	1 cm = 125 ml	200 units = 3.2 cm

Rig time runs 10 times slower than the program at speed ×1: the “t = … s” line in the program is model seconds, multiply them by 10.

Parts

#	Part	Qty	Notes
1	Bottom tank, 3 L container	1	base 20 × 10 cm (200 cm²), ≥ 15 cm tall
2	Tank, 2 L container	3	base 12.5 × 10 cm, scale on the side
3	Submersible 12 V pump	1	brushless, 240–300 L/h, head ≥ 3 m
4	Splitter t (trough)	1	60 mm weir, one plate
5	Splitter s, m (trough)	1	60 mm weir, two plates
6	Bulkhead fitting 8 mm	3	aquarium type, with gaskets
7	Aquarium valve 8 mm	3	G, I, X; 40 cm below tank bottoms
8	Silicone tube 8 mm, tees	~5 m	food grade, soft
9	HC-SR04 sensor	1	above the bottom tank, looking down
10	Arduino Nano + MOSFET	1	IRLZ44N (logic level), 1N4007 diode across the pump
11	12 V 2 A power supply	1	enclosed, away from water
12	Plywood board 80 × 75 cm	1	shelf, feet, cable ties, clamps

Also: food colouring, a 250 ml measuring cup, a stopwatch (phone), a spirit level, a sponge.

Splitters t, s, m

This is the key trick of the Phillips machine. Water spills over a straight horizontal edge 60 mm wide, and a vertical plate divides the edge: the share of the flow = the share of the width. Make the trough from a food container: cut a window in the wall and glue a straight plastic strip along its bottom — the weir. 1 mm plastic plates slide in slots; behind the weir each part drains into its own hose.

The weir must be perfectly level (check with a spirit level), and the water in front of it must be calm: put a piece of sponge under the jet from the hose.

Lever	Plate from the left edge	Default	Presets
t	t · 60 mm	15.0 mm	stimulus 10.8, austerity 22.8
s	first plate: s · 60 mm	9.0 mm	recession 19.2
m	second plate: another m · 60 mm	9.0 mm	—

Valves G, I, X: nearly constant outflow

In the model spending, investment and exports do not depend on the tank level. On the rig this is done with height: each valve sits 40 cm below its tank bottom, so the head only changes from 40 to 56 cm. Flow grows as the square root of head, so with a full tank it is only 1.18 times the flow of an empty one.

Set a valve with the tank half full: hold the measuring cup under it for 15 seconds and turn until it collects the volume from the table.

Valve	Model value	ml/min	ml in 15 s
G	G = 20	240	60
G	G = 50	600	150
G	G = 85	1020	255
X	X = 8	96	24
X	X = 22	264	66
X	X = 40	480	120
I	r = 0.0% → I = 45.0	540	135
I	r = 1.5% → I = 35.9	431	108
I	r = 4.0% → I = 24.7	296	74
I	r = 9.0% → I = 11.7	140	35
I	r = 12.0% → I = 7.4	89	22

On the rig the rate r is “turned” with valve I: mark its positions for different r using this table.

Pump: output proportional to the water in the bottom tank

An ordinary pump gives a nearly constant flow. For the multiplier to appear, output must grow with the bottom tank level, as in the model: Y = 0.5 · level. Arduino does this: the HC-SR04 measures the level and the MOSFET changes the pump PWM. On the rig Y [ml/s] = 0.05 · water [ml] — e.g. about 147 L/h with the default levers.

The sketch is **phillips_pump.ino**: in the program archive (phillips_pump folder), on GitHub — github.com/avv123avv/phillips_machine/blob/main/diy/phillips_pump/phillips_pump.ino, full text in the appendix at the end of this manual. Once a second it writes t_s, level_cm, water_ml, y_ml_s, pwm to the serial port — open it in the Arduino IDE Serial Plotter. A button on D2 turns the pump on and off.

There must always be a 500 ml “dead” reserve under the pump, otherwise it sucks air. It is not part of the model: the sketch subtracts it from the volume.

Safety. Only 12 V near water; the power supply enclosed and above the water level; the diode across the pump is a must.

Calibration and first run

1. Assemble everything dry, level the splitter weirs. Make sure no hose rises above its tank — otherwise you get a siphon.

2. Pump: temporarily set PWM to 80, 120, 160, 200, 255 in the sketch and measure the output with the cup at board height. Put PUMP_MAX_ML_S (output at 255) and PWM_MIN (where the pump starts pumping) into the sketch.

3. Sensor: measure the distance from the sensor to the bottom tank's floor and put it into SENSOR_TO_BOTTOM_CM. Pour in 1 L — the serial port should show level_cm ≈ 5.

4. Fill as after a Reset: bottom tank 1300 ml (with the reserve), Treasury, idle and foreign balances 400 ml each. Total 2.5 L. Colour the water.

5. Set the plates and valves from the tables for the default levers, start the pump and wait 10 minutes — levels should nearly stop changing (60 s in the program).

6. Now repeat the experiments and problems from sections 6 and 7, comparing the tank levels with the “Build scheme” screen. A 10–20% difference is normal: splitters and valves are not perfect.

Appendix. Arduino pump sketch

File phillips_pump.ino is in the program archive and on GitHub: github.com/avv123avv/phillips_machine/blob/main/diy/phillips_pump/phillips_pump.ino. In the Arduino IDE the file must be in a folder with the same name: phillips_pump/phillips_pump.ino. Before uploading, set SENSOR_TO_BOTTOM_CM, PUMP_MAX_ML_S and PWM_MIN (section “Calibration and first run”). Comments in the sketch are in Russian.

```
// Насос домашней машины Филлипса: подача помпы пропорциональна объёму воды в нижнем баке,
// как в модели программы: Y = PUMP_K · уровень (economy.gd), с поправкой на масштаб времени стенда.
//
// Детали: Arduino Nano, ультразвуковой датчик HC-SR04 над нижним баком (смотрит вниз на воду),
// MOSFET-модуль (IRF520 / IRLZ44N с логическим затвором) в минусе помпы 12 В, диод на помпе,
// кнопка вкл/выкл насоса на D2 (на землю). Земли Arduino и блока 12 В соединены.
//
// Порт: 115200 бод, раз в секунду строка CSV: t_s,level_cm,water_ml,y_ml_s,pwm
// Калибровка — раздел «Сборка дома» в инструкции (F1 в программе).

// --- Подключение ------------------------------------------------------------
const int PIN_TRIG = 9;
const int PIN_ECHO = 10;
const int PIN_PUMP = 3;       // ШИМ
const int PIN_BUTTON = 2;

// --- Геометрия стенда (как в diy_rig.gd) ------------------------------------
const float SENSOR_TO_BOTTOM_CM = 20.0;  // от датчика до дна нижнего бака — измерьте линейкой
const float SUMP_AREA_CM2 = 200.0;       // дно 20 × 10 см
const float SUMP_DEAD_ML = 500.0;        // запас под помпой, в модель не входит

// --- Модель -----------------------------------------------------------------
// PUMP_K = 0,5 1/с модели, 1 с модели = 10 с стенда → 0,05 1/с стенда.
const float PUMP_K_PER_S = 0.05;

// --- Калибровка помпы -------------------------------------------------------
// PUMP_MAX_ML_S — подача при ШИМ 255 на высоту вашего щита (240 л/ч ≈ 66,7 мл/с по паспорту,
// реально меньше — измерьте мерным стаканом). PWM_MIN — ниже этого ШИМ помпа не крутится.
```

```
const float PUMP_MAX_ML_S = 66.7;
const int PWM_MIN = 60;

bool pumpOn = true;
float levelCm = 0.0;
unsigned long lastReport = 0;

float readDistanceCm() {
  digitalWrite(PIN_TRIG, LOW);
  delayMicroseconds(2);
  digitalWrite(PIN_TRIG, HIGH);
  delayMicroseconds(10);
  digitalWrite(PIN_TRIG, LOW);
  unsigned long us = pulseIn(PIN_ECHO, HIGH, 30000UL);
  if (us == 0) return -1.0;
  return us * 0.0343 / 2.0;  // скорость звука ~343 м/с при 20 °C
}

// Медиана из пяти замеров — рябь на воде и отражения дают выбросы.
float readLevelCm() {
  float v[5];
  int n = 0;
  for (int i = 0; i < 5; i++) {
    float d = readDistanceCm();
    if (d > 0) v[n++] = d;
    delay(12);
  }
  if (n == 0) return levelCm;
  for (int i = 0; i < n; i++)
    for (int j = i + 1; j < n; j++)
      if (v[j] < v[i]) { float t = v[i]; v[i] = v[j]; v[j] = t; }
  return SENSOR_TO_BOTTOM_CM - v[n / 2];
}

void setup() {
  pinMode(PIN_TRIG, OUTPUT);
  pinMode(PIN_ECHO, INPUT);
  pinMode(PIN_PUMP, OUTPUT);
  pinMode(PIN_BUTTON, INPUT_PULLUP);
  Serial.begin(115200);
```

```
  Serial.println("t_s,level_cm,water_ml,y_ml_s,pwm");
  levelCm = readLevelCm();
}

void loop() {
  static bool lastButton = HIGH;
  bool button = digitalRead(PIN_BUTTON);
  if (lastButton == HIGH && button == LOW) pumpOn = !pumpOn;
  lastButton = button;

  // Сглаживание: уровень меняется медленно, датчик шумит.
  levelCm += 0.3 * (readLevelCm() - levelCm);

  // Вода в обороте = объём над мёртвым запасом; Y = K · вода.
  float waterMl = levelCm * SUMP_AREA_CM2 - SUMP_DEAD_ML;
  if (waterMl < 0) waterMl = 0;
  float yMlS = pumpOn ? PUMP_K_PER_S * waterMl : 0.0;
  int pwm = 0;
  if (yMlS > 0.5) {
    float share = yMlS / PUMP_MAX_ML_S;
    if (share > 1) share = 1;
    pwm = PWM_MIN + (int)((255 - PWM_MIN) * share);
  }
  analogWrite(PIN_PUMP, pwm);

  unsigned long now = millis();
  if (now - lastReport >= 1000) {
    lastReport = now;
    Serial.print(now / 1000.0, 1); Serial.print(',');
    Serial.print(levelCm, 2); Serial.print(',');
    Serial.print(waterMl, 0); Serial.print(',');
    Serial.print(yMlS, 1); Serial.print(',');
    Serial.println(pwm);
  }
}
```